GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

MA.8.GR.2.4

I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

Warm-Up

1. The triangle $(0, 0)$, $(3, 0)$, $(3, 2)$: its perimeter, to the tenth?
2. A 13-ft ladder's foot is 5 ft from a wall. How high does it reach?
3. A 10 cm side is 4 cm in a copy. What is the scale factor?
4. At 1 in : 6 ft, a 3.5-in wall on a drawing is how long?

Warm-Up

1. The triangle $(0, 0)$, $(3, 0)$, $(3, 2)$: its perimeter, to the tenth?

   8.6 units

2. A 13-ft ladder's foot is 5 ft from a wall. How high does it reach?

   12 ft

3. A 10 cm side is 4 cm in a copy. What is the scale factor?

   k = 0.4

4. At 1 in : 6 ft, a 3.5-in wall on a drawing is how long? **21 ft**

Vocabulary

similar triangles

the same shape, any size: matching angles equal, and every side multiplied by the same number

corresponding sides

sides that match — across from equal angles: shortest with shortest, longest with longest

scale factor k

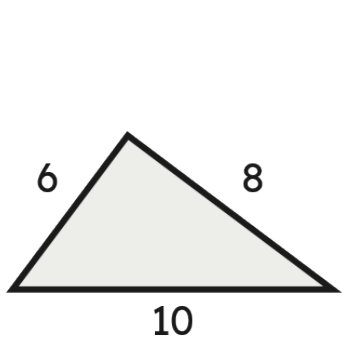
copy ÷ original, for one matching pair of sides

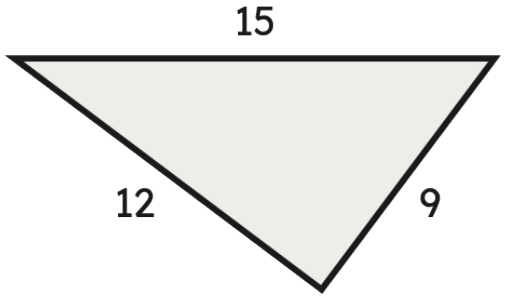
indirect measurement

finding a length you cannot reach from similar triangles you can measure

Notes

I. Similar: every side times the same k



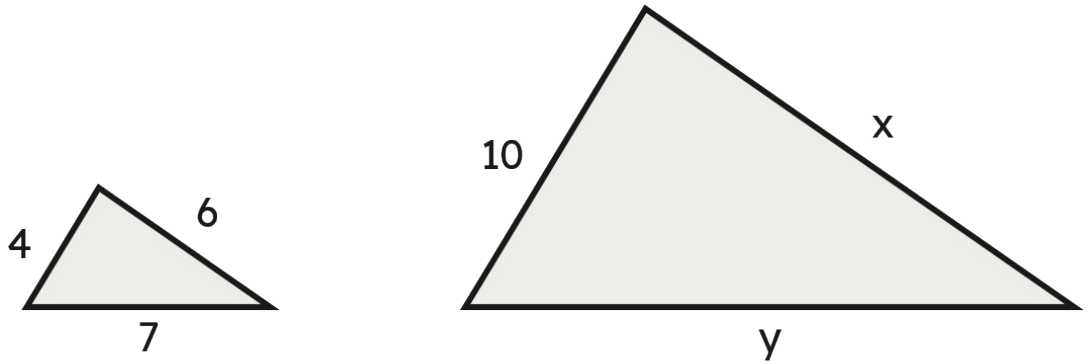


$$\frac{9}{6} = 1.5, \quad \frac{12}{8} = 1.5, \quad \frac{15}{10} = 1.5$$

Three ratios, all equal: similar. Match shortest to shortest, longest to longest.

Notes

II. A missing side: find k, then multiply

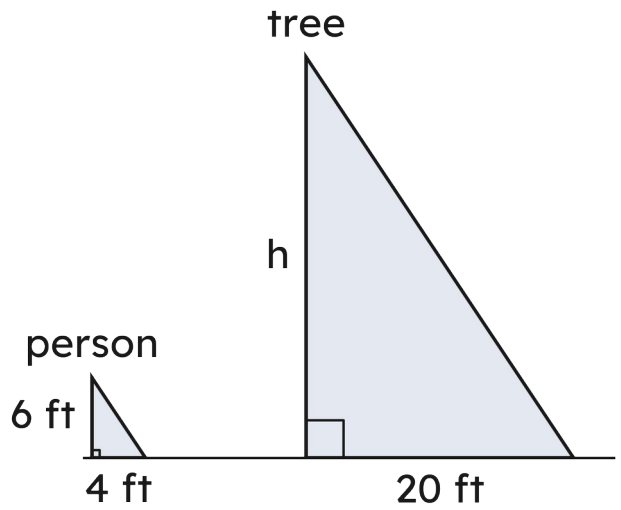


$$k = \frac{10}{4} = 2.5$$

$$x = 6 \times 2.5 = 15, \quad y = 7 \times 2.5 = 17.5$$

Notes

III. Same sun, same angle: shadows

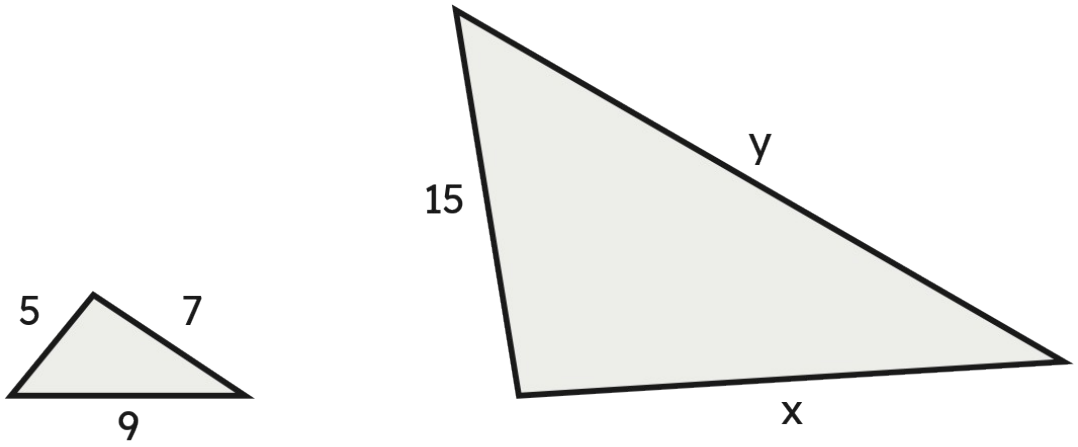


$$k = \frac{20}{4} = 5$$

$$h = 6 \times 5 = 30 \text{ ft}$$

Example 1

The two triangles are similar.



Find x and y.

Example 1

$$k = \frac{15}{5} = 3$$

$$x = 7 \times 3 = 21$$

$$y = 9 \times 3 = 27$$

Answer: x = 21, y = 27

Your Turn

A triangle has sides 3, 5 and 7.

A similar triangle's shortest side is 12. Find its other two sides.

Your Turn

A triangle has sides 3, 5 and 7.

A similar triangle's shortest side is 12. Find its other two sides.

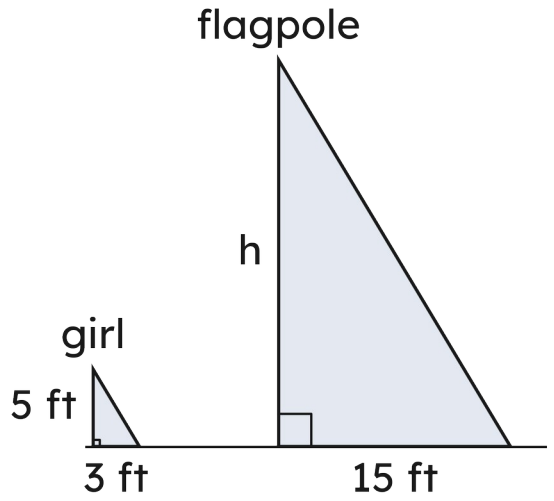
$$k = \frac{12}{3} = 4$$

$$5 \times 4 = 20, \quad 7 \times 4 = 28$$

Answer: 20 and 28

Example 2

A 5-ft girl casts a 3-ft shadow. At the same moment, a flagpole casts a 15-ft shadow.



How tall is the flagpole?

Example 2

$$k = \frac{15}{3} = 5$$

$$h = 5 \times 5 = 25$$

Answer: The flagpole is 25 ft tall.

Your Turn

A 2-m post casts a 1.5-m shadow. At the same moment, a building casts a 12-m shadow. How tall is the building?

Your Turn

A 2-m post casts a 1.5-m shadow. At the same moment, a building casts a 12-m shadow. How tall is the building?

$$k = \frac{12}{1.5} = 8$$

$$h = 2 \times 8 = 16$$

Answer: 16 m

Whiteboards · Question 1 of 9

Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?

Answer it.

Whiteboards · Question 1 of 9

Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?

$$\frac{12}{8} = 1.5, \quad \frac{18}{12} = 1.5, \quad \frac{21}{14} = 1.5$$

all three equal

Answer: Yes — every ratio is 1.5

Whiteboards · Question 2 of 9

A triangle has sides 6 cm, 8 cm and 11 cm.

A similar triangle's shortest side is 18 cm.

Find its other two sides.

Answer it.

Whiteboards · Question 2 of 9

A triangle has sides 6 cm, 8 cm and 11 cm.

A similar triangle's shortest side is 18 cm.

Find its other two sides.

$$k = \frac{18}{6} = 3$$

$$8 \times 3 = 24, \quad 11 \times 3 = 33$$

Answer: 24 cm and 33 cm

Whiteboards · Question 3 of 9

A triangle has sides 12 in, 16 in and 20 in.

A similar triangle's longest side is 5 in.

Find its other two sides.

Answer it.

Whiteboards · Question 3 of 9

A triangle has sides 12 in, 16 in and 20 in.

A similar triangle's longest side is 5 in.

Find its other two sides.

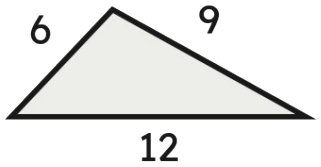
$$k = \frac{5}{20} = 0.25$$

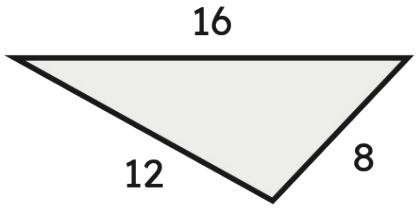
$$12 \times 0.25 = 3, \quad 16 \times 0.25 = 4$$

Answer: 3 in and 4 in

Whiteboards · Question 4 of 9

Similar triangles: which side of the second matches the 9?





A. 8

B. 16

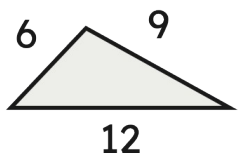
C. 12

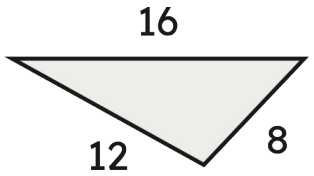
D. 9

Answer it.

Whiteboards · Question 4 of 9

Similar triangles: which side of the second matches the 9?





sorted: 6, 9, 12 and 8, 12, 16

the middle sides match: 9 and 12

Answer: C — the 12

Whiteboards · Question 5 of 9

Which triangle is similar to one with sides 5, 6 and 8?

A. 7, 8, 10

B. 10, 12, 16

C. 10, 12, 15

D. 5, 6, 9

Answer it.

Whiteboards · Question 5 of 9

Which triangle is similar to one with sides 5, 6 and 8?

$$\frac{10}{5} = 2, \quad \frac{12}{6} = 2, \quad \frac{16}{8} = 2$$

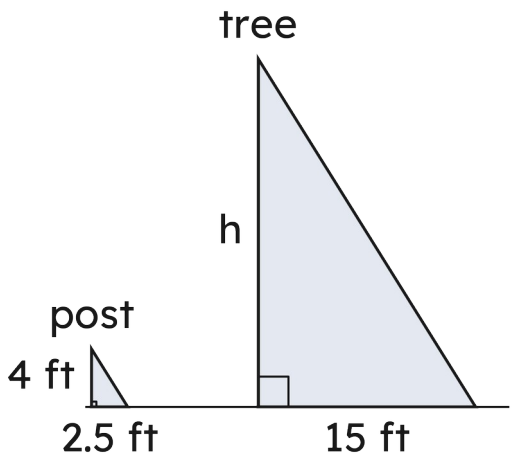
one k for every side

Answer: B — 10, 12, 16

Whiteboards · Question 6 of 9

A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow.

How tall is the tree?

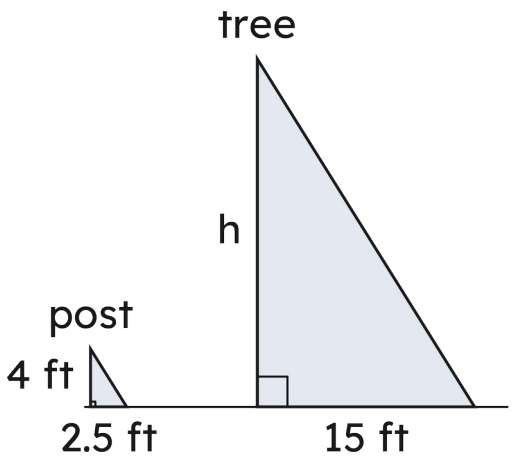


Answer it.

Whiteboards · Question 6 of 9

A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow.

How tall is the tree?



$$k = \frac{15}{2.5} = 6$$

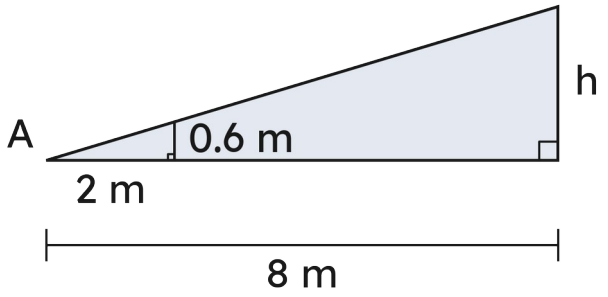
$$h = 4 \times 6 = 24$$

Answer: 24 ft

Whiteboards · Question 7 of 9

A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar.

How high is the ramp 8 m from A?

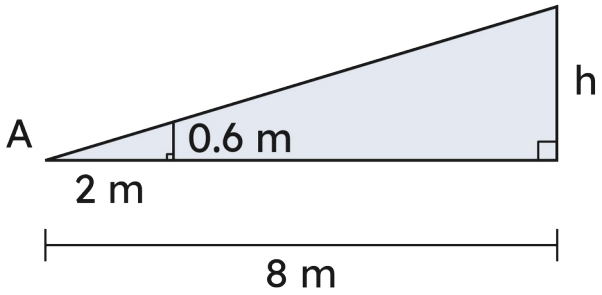


Answer it.

Whiteboards · Question 7 of 9

A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar.

How high is the ramp 8 m from A?



$$k = \frac{8}{2} = 4$$

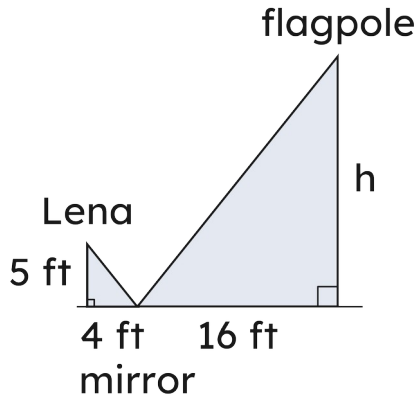
$$h = 0.6 \times 4 = 2.4$$

Answer: 2.4 m

Whiteboards · Question 8 of 9

Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror.

The triangles are similar. How tall is the flagpole?

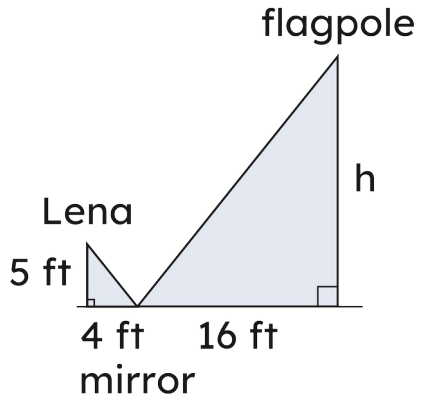


Answer it.

Whiteboards · Question 8 of 9

Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror.

The triangles are similar. How tall is the flagpole?



$$k = \frac{16}{4} = 4$$

$$h = 5 \times 4 = 20$$

Answer: 20 ft

Whiteboards · Question 9 of 9

Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2.

Is she right? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2.

Is she right? Explain.

$\frac{5}{3} \approx 1.67$, $\frac{6}{4} = 1.5$, $\frac{7}{5} = 1.4$: three ratios

Answer: No — similar means × one k, not + 2.

Independent Set

1. Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

2. A triangle has sides 8, 10 and 13. A similar triangle's 10-side is 25. Find its other two sides.

3. A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

4. A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

5. Two similar right triangles share corner A. The small one runs 3 m along the ground and stands 1.2 m high. The big one's upright is 9 m beyond the small one's. How high is it?

6. Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Side lengths of similar triangles — QSK

Similar triangles and indirect measurement — 88W

Identify similar triangles — 6PD